

Mini Open Methods Memo (Activity #14)

Sumaiya Azad

2025-10-20

1 Introduction

This memo brings together several key activities from the course into one reproducible document. I revisit data wrangling with the Armed Forces data, extend my baby names visualization, and return to the Box Problem from early in the semester. I end with a brief reflection on what I feel I have learned so far. All of the code used to generate the results in this document is collected in a Code Appendix at the end.

2 Armed Forces Data Wrangling

In Activities #08 and #10, we used the Armed Forces data set to practice data wrangling and create frequency tables. Here I present a revised and fully reproducible version of that work. Instead of reading a local Excel file, I read directly from a Google Sheets URL so that the code can run on someone else's computer without modification.

For this memo, I focus on a narrower sub-set of soldiers:

- **Sex:** Female soldiers only
- **Branches:** All branches represented in the table
- **Case:** Each row in the expanded data set represents an individual soldier

The original Excel file has multiple header rows with merged cells. I reconstruct the header by filling in missing branch names and combining branch and sex labels into meaningful column names. I then pivot the data into a tidy format, convert counts to numeric values, and use `uncount()` to expand the table so that each row corresponds to an individual soldier. Finally, I create a two-way frequency table of **branch** by **rank** for female soldiers, with both counts and percentages.

Table 1: Female soldiers by branch and rank

branch	E1	E2	E3	E4	E5	E6	E7	E8	E9	O1	O2	O3	O4	O5	O6	O7	O8	O9	Total	Total (count)	Total Officers	Total Warrant Officers	W1	W2	W3	W4	W5
Army	1,234 (0.1%)	2,345 (0.2%)	3,456 (0.3%)	4,567 (0.4%)	5,678 (0.5%)	6,789 (0.6%)	7,890 (0.7%)	8,901 (0.8%)	9,012 (0.9%)	10,123 (1.0%)	11,234 (1.1%)	12,345 (1.2%)	13,456 (1.3%)	14,567 (1.4%)	15,678 (1.5%)	16,789 (1.6%)	17,890 (1.7%)	18,901 (1.8%)	19,012 (1.9%)	20,123 (2.0%)	21,234 (2.1%)	22,345 (2.2%)	23,456 (2.3%)	24,567 (2.4%)	25,678 (2.5%)	26,789 (2.6%)	27,890 (2.7%)
Navy	1,234 (0.1%)	2,345 (0.2%)	3,456 (0.3%)	4,567 (0.4%)	5,678 (0.5%)	6,789 (0.6%)	7,890 (0.7%)	8,901 (0.8%)	9,012 (0.9%)	10,123 (1.0%)	11,234 (1.1%)	12,345 (1.2%)	13,456 (1.3%)	14,567 (1.4%)	15,678 (1.5%)	16,789 (1.6%)	17,890 (1.7%)	18,901 (1.8%)	19,012 (1.9%)	20,123 (2.0%)	21,234 (2.1%)	22,345 (2.2%)	23,456 (2.3%)	24,567 (2.4%)	25,678 (2.5%)	26,789 (2.6%)	27,890 (2.7%)
Air Force	1,234 (0.1%)	2,345 (0.2%)	3,456 (0.3%)	4,567 (0.4%)	5,678 (0.5%)	6,789 (0.6%)	7,890 (0.7%)	8,901 (0.8%)	9,012 (0.9%)	10,123 (1.0%)	11,234 (1.1%)	12,345 (1.2%)	13,456 (1.3%)	14,567 (1.4%)	15,678 (1.5%)	16,789 (1.6%)	17,890 (1.7%)	18,901 (1.8%)	19,012 (1.9%)	20,123 (2.0%)	21,234 (2.1%)	22,345 (2.2%)	23,456 (2.3%)	24,567 (2.4%)	25,678 (2.5%)	26,789 (2.6%)	27,890 (2.7%)
Marine Corps	1,234 (0.1%)	2,345 (0.2%)	3,456 (0.3%)	4,567 (0.4%)	5,678 (0.5%)	6,789 (0.6%)	7,890 (0.7%)	8,901 (0.8%)	9,012 (0.9%)	10,123 (1.0%)	11,234 (1.1%)	12,345 (1.2%)	13,456 (1.3%)	14,567 (1.4%)	15,678 (1.5%)	16,789 (1.6%)	17,890 (1.7%)	18,901 (1.8%)	19,012 (1.9%)	20,123 (2.0%)	21,234 (2.1%)	22,345 (2.2%)	23,456 (2.3%)	24,567 (2.4%)	25,678 (2.5%)	26,789 (2.6%)	27,890 (2.7%)
Cosmos	1,234 (0.1%)	2,345 (0.2%)	3,456 (0.3%)	4,567 (0.4%)	5,678 (0.5%)	6,789 (0.6%)	7,890 (0.7%)	8,901 (0.8%)	9,012 (0.9%)	10,123 (1.0%)	11,234 (1.1%)	12,345 (1.2%)	13,456 (1.3%)	14,567 (1.4%)	15,678 (1.5%)	16,789 (1.6%)	17,890 (1.7%)	18,901 (1.8%)	19,012 (1.9%)	20,123 (2.0%)	21,234 (2.1%)	22,345 (2.2%)	23,456 (2.3%)	24,567 (2.4%)	25,678 (2.5%)	26,789 (2.6%)	27,890 (2.7%)
Total	1,234 (0.1%)	2,345 (0.2%)	3,456 (0.3%)	4,567 (0.4%)	5,678 (0.5%)	6,789 (0.6%)	7,890 (0.7%)	8,901 (0.8%)	9,012 (0.9%)	10,123 (1.0%)	11,234 (1.1%)	12,345 (1.2%)	13,456 (1.3%)	14,567 (1.4%)	15,678 (1.5%)	16,789 (1.6%)	17,890 (1.7%)	18,901 (1.8%)	19,012 (1.9%)	20,123 (2.0%)	21,234 (2.1%)	22,345 (2.2%)	23,456 (2.3%)	24,567 (2.4%)	25,678 (2.5%)	26,789 (2.6%)	27,890 (2.7%)

Table 1 summarizes the distribution of female soldiers by branch and rank. Each cell contains the number of soldiers and the corresponding overall percentage. The totals row and column provide

convenient summaries across all branches and all ranks. We can see that some branches have a higher number of female soldiers overall, and within those branches certain ranks are more common than others. This suggests that the relationship between branch and rank is not uniform even within the female sub-group, and the table helps identify which branch–rank combinations are most prevalent.

3 Popularity of Baby Names over the Years

The BabyNames data set from the dcData package contains counts of how many babies received each name by year and sex in the United States. For this memo I chose to highlight the **top three female names** and **top three male names** based on total counts across all available years.

I first identify the top three names for each sex, then create a data set that contains only those names. In the plot, I distinguish sexes using **line type**: dashed lines for girls and solid lines for boys. Using both color and line type helps make the figure more accessible.

Figure 1: Figure 1. Popularity of the top three girl and boy names in the U.S. over time.

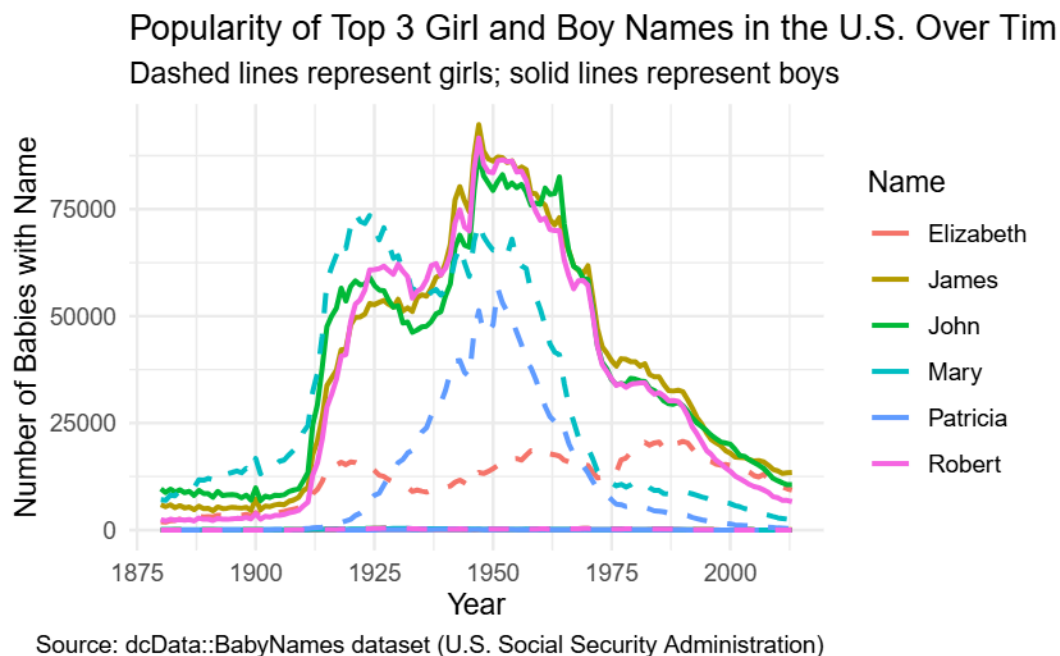


Figure 1 shows how the popularity of the top names for girls and boys changes over time. Some names dominate early decades and then decline, while others become more popular only in recent years. The dashed versus solid line types make it easy to see differences between female and male usage. I chose to use the top three names for each sex because they provide clear trends in the data and highlight how naming patterns evolve across the years.

4 Plotting a Mathematical Function: The Box Problem

In the Box Problem from Activity #04, we start with a rectangular piece of paper, cut out equal squares of side length (x) from each corner, and fold up the resulting flaps to form an open-top box. If the original sheet has length (L) and width (W), then the volume of the resulting box is

$$[V(x) = x \times (L - 2x) \times (W - 2x).]$$

For this memo I use a sheet that is **36 inches by 48 inches**, taking ($L = 48$) inches and ($W = 36$) inches. The function below implements this formula, and I use `{ggplot2}` with `stat_function()` to plot the volume as a function of the cut size (x).

Figure 2: Figure 2. Volume of an open-top box made from a 36 by 48 inch sheet as a function of cut size.

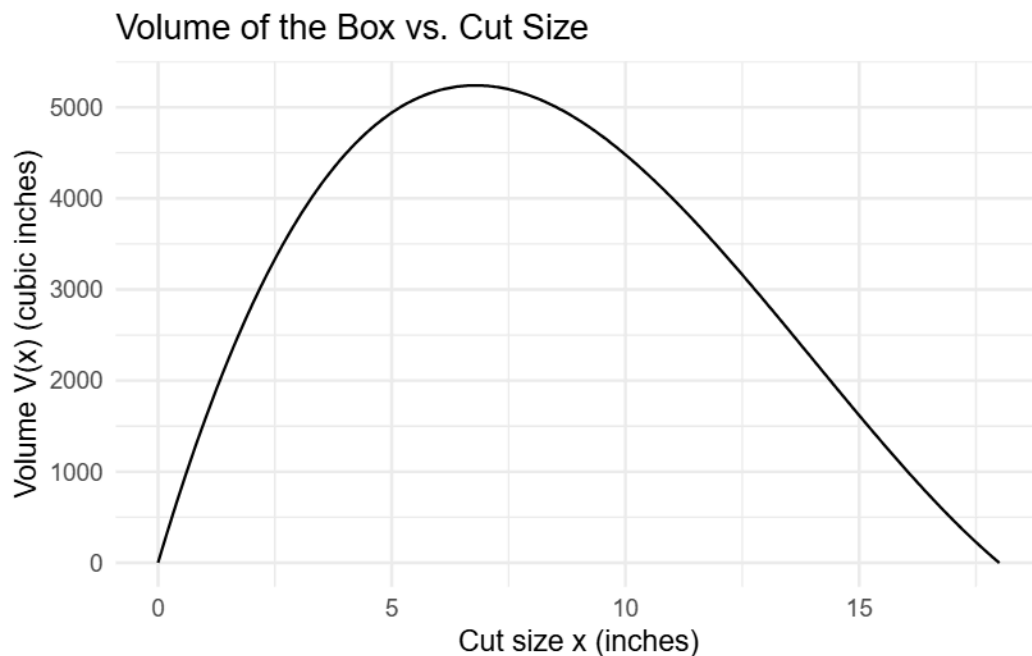


Figure 2 shows that the volume starts at zero when ($x = 0$), increases to a single peak, and then decreases back toward zero as the cut size becomes too large. This pattern reflects the trade-off between increasing the height of the box (with larger cuts) and shrinking the base dimensions ($(L - 2x)$) and $(W - 2x)$). The maximum volume occurs at an intermediate cut size (approximately 6–7 inches), which yields a volume of around 5,000–5,300 cubic inches. In words, cutting out squares of about 6–7 inches from each corner of the 36 by 48 inch sheet produces the box with the largest possible volume.

5 What I Feel I've Learned So Far

So far in this course I feel that I have:

- **Learned how to write reproducible code** that can be run on someone else's computer. Moving my Armed Forces wrangling code away from a local file path and reading directly from a shared Google Sheet was a concrete step toward reproducibility.
- **Strengthened my data wrangling skills**, especially with `pivot_longer()`, `filter()`, and the `{janitor}` functions such as `tabyl()` and `adorn_*()`. Working through the messy Armed Forces headers made me more comfortable with real-world, imperfect data.
- **Improved my visualization practice** by using `{ggplot2}` and thinking about effective graphics. In the baby names plot, I used line type in addition to color to distinguish sexes, which helps with accessibility and avoids relying on color alone.

Overall, I feel much more comfortable with R syntax and with the full data analysis cycle. The Plan–Code–Improve–Polish framework has helped me structure my workflow more intentionally. Over time I have gradually gained more confidence in my data wrangling and visualization skills, and I'm working towards becoming more familiar with producing and publishing reproducible work.

6 Code Appendix

All R code used to generate the tables and figures in this document is collected here.

```
# ---- Setup & packages -----

library(readxl)
library(dplyr)
library(tidyr)
library(janitor)
library(knitr)
library(kableExtra)
library(dcData)
library(ggplot2)

knitr::opts_chunk$set(
  echo      = FALSE,
  message   = FALSE,
  warning    = FALSE
)

# ---- Armed Forces: wrangle & frequency table (female soldiers) -----

sheet_url <- "https://docs.google.com/spreadsheets/d/1cn4i0-ymB1ZytWXCwsJiq6fZ9P
hGLUvbMBHlzqG4bwo/export?format=xlsx&gid=597536282"
temp_xlsx <- tempfile(fileext = ".xlsx")
download.file(sheet_url, destfile = temp_xlsx, mode = "wb")

df_raw <- read_excel(temp_xlsx, col_names = FALSE)

header1 <- as.character(unlist(df_raw[2, ]))
```

```

header2 <- as.character(unlist(df_raw[3, ]))

last_branch <- NA_character_
for (i in seq_along(header1)) {
  if (!is.na(header1[i]) && nzchar(header1[i])) {
    last_branch <- header1[i]
  } else {
    header1[i] <- last_branch
  }
}

col_names <- ifelse(
  is.na(header2) | !nzchar(header2),
  header1,
  paste(header1, header2)
)

data_clean <- df_raw[-c(1:3), ]
colnames(data_clean) <- col_names
colnames(data_clean)[1] <- "pay_grade"

Armed_Forces <- data_clean |>
  pivot_longer(
    cols      = -pay_grade,
    names_to  = c("branch", "sex"),
    names_sep = " ",
    values_to = "count"
  ) |>
  filter(!is.na(count)) |>
  mutate(branch = trimws(branch)) |>
  filter(sex != "Total") |>
  mutate(count = as.numeric(count)) |>
  filter(!is.na(count))

Armed_Forces_Individuals <- Armed_Forces |>
  uncount(weights = count, .remove = FALSE) |>
  mutate(soldier_id = dplyr::row_number()) |>
  select(soldier_id, pay_grade, branch, sex) |>
  rename(rank = pay_grade)

make_table <- function(data, sex_label) {
  data |>
    filter(sex == sex_label) |>
    tabyl(branch, rank) |>
    adorn_totals(where = c("row", "col")) |>
    adorn_percentages("all") |>
    adorn_pct_formatting(digits = 1) |>

```

```

    adorn_ns(position = "front") |>
    kbl(
      caption = paste(sex_label, "soldiers by branch and rank"),
      align    = "c",
      booktabs = TRUE
    ) |>
    kable_styling(
      position      = "center",
      latex_options = c("hold_position", "scale_down")
    )
  }

female_table <- make_table(Armed_Forces_Individuals, "Female")

# ---- Popular Baby Names: top 3 girl & boy names -----

data("BabyNames")

top_female <- BabyNames |>
  filter(sex == "F") |>
  group_by(name) |>
  summarise(total = sum(count, na.rm = TRUE), .groups = "drop") |>
  arrange(desc(total)) |>
  slice_head(n = 3) |>
  pull(name)

top_male <- BabyNames |>
  filter(sex == "M") |>
  group_by(name) |>
  summarise(total = sum(count, na.rm = TRUE), .groups = "drop") |>
  arrange(desc(total)) |>
  slice_head(n = 3) |>
  pull(name)

baby_data <- BabyNames |>
  filter(name %in% c(top_female, top_male)) |>
  mutate(line_type = ifelse(sex == "F", "dashed", "solid"))

ggplot(
  data      = baby_data,
  mapping = aes(
    x      = year,
    y      = count,
    colour = name,
    linetype = line_type
  )
) +

```

```

geom_line(linewidth = 0.9) +
scale_linetype_identity() +
labs(
  title    = "Popularity of Top 3 Girl and Boy Names in the U.S. Over Time",
  subtitle = "Dashed lines represent girls; solid lines represent boys",
  x        = "Year",
  y        = "Number of Babies with Name",
  colour   = "Name",
  caption  = "Source: dcData::BabyNames dataset (U.S. Social Security
  Administration)"
) +
theme_minimal()

# ---- Box Problem: function & stat_function plot -----

volume_box <- function(x, length_in = 48, width_in = 36) {
  x * (length_in - 2 * x) * (width_in - 2 * x)
}

box_domain <- data.frame(
  x = c(0, 18)
)

ggplot(
  data      = box_domain,
  mapping   = aes(x = x)
) +
  stat_function(
    fun = volume_box,
    n   = 501
  ) +
  labs(
    title = "Volume of the Box vs. Cut Size",
    x     = "Cut size x (inches)",
    y     = "Volume V(x) (cubic inches)"
  ) +
  theme_minimal()

```